



QUICKMATHS / STUDENT EDITION

The complete student guide

Learn deliberately. Show your reasoning.
Keep your progress yours.

LEARNER REFERENCE

Every visible learner control, study workflow, review state, safety boundary, and recovery path.

AGENT START

Get the QuickMaths agent guide summary, check my app state and progress, then guide me through the learning experience.

Version 21 / September 2026



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A complete reference to the learner workspace, study workflow, and portable progress model.

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1. How QuickMaths works

QuickMaths connects lessons into mastery maps instead of treating every topic as an isolated quiz. A lesson can recommend or require earlier lessons, unlock later work, and bridge into another subject.

Layer	What it contains	What it means for you
Profile	Progress, attempts, reviews, drafts, timers, personal plans, and preferences	Your learning record stays separate from other people using the browser
Subject	Lessons, prerequisite graph, theme, and cross-subject bridges	Switching subject changes the visible curriculum and color identity
Curriculum	An educator-authored selection of packs, canonical map, learning path, and optional agent policy	A portable learning plan can travel from an educator to your profile
Lesson pack	One subject or set of lessons, questions, grading rules, and work requirements	Installed packs extend the same map, progress, testing, and backup system

Mastery is more than a score

QuickMaths records final-answer grading, required reasoning, reflection, review status, confidence, mistake tags, and practice history. Formal proofs and rubric responses can stay pending until an allowed reviewer passes them. A correct final conclusion alone does not prove that the reasoning is valid.

Local-first ownership

Meaningful changes autosave in this browser. No QuickMaths account or model API key is required. Browser autosave is convenient, but it belongs to this site origin on this device. Use:

- a full JSON backup for complete recovery or device migration;
- optional GitHub Bridge storage for persistent cross-device checkpoints;
- curriculum files to load an educator-authored learning plan.

CSV files are for analysis. They cannot restore your profile.

2. Landing page and learner profiles

The landing page offers **I'm learning** and **I'm designing a curriculum**. Learners normally use the first path.

Landing header and live counts

The QuickMaths brand returns to the profile picker. The local-first pill summarizes the account-free core. Live counts report connected lessons, varied mastery questions, and registered WebMCP actions from the current build.

Choose or create a learner

Existing profile cards show the learner name and accumulated practice time. Select one to continue. A new profile needs only a display name. Profiles do not create online accounts.

New learner profiles begin with the seven-chapter app tutorial. You can skip it and replay it later from Settings without resetting progress.

Load a curriculum

Use a local curriculum JSON file or a public GitHub file URL. QuickMaths previews the curriculum before attaching it. A curriculum can carry:

- enabled additive lesson packs;
- a canonical map arrangement;
- educator-created paths and annotations;
- Hard or Open learning path;
- optional student name and proof contact;
- private agent instructions and Agent in the loop status.

If you load a curriculum before choosing a profile, QuickMaths holds it for the next learner you create or select.

Restore from GitHub storage

Already have a profile on another device? opens the storage connection form. Enter the repository owner, repository, branch, and fine-grained token privately in the app. QuickMaths can then load the complete learner checkpoint instead of starting over.

Load backup and sample progress

Load backup previews a complete QuickMaths JSON backup before replacing local state. **Explore with sample progress** creates a disposable learner with example progress so you can examine the interface.

Student guide link

The learner landing panel links to this PDF. The same guide remains available from the learner dashboard and Settings.

3. Seven-chapter app tutorial

The tutorial is a visual introduction, not a separate account setup.

Chapter	What it demonstrates
1. Your workspace	Local profile, autosave, portable backup, and account-free use
2. Subject and path	Subject picker, Hard path, and Open path; you can choose the mode when no curriculum controls it
3. Mastery map	Prerequisites, status, map movement, zoom, and personal Plan mode
4. Learning loop	Theory, examples, complete authored tests, shown work, reflection, and mastery
5. Lesson ecosystem	Lesson Depot discovery and Lesson Studio authoring or native improvement

Chapter	What it demonstrates
6. Agent support	WebMCP connection, suggested tutor prompt, visible agent actions, and answer-key boundary
7. Ownership	Browser autosave, JSON backup, storage bridge, and portability

Skip tour moves to the dashboard. **Back** and **Next** move between chapters. The final chapter opens the app, and Lesson Studio can be opened directly from its tutorial chapter.

4. Shared application shell

The shell keeps navigation, subject, identity, time, and agent status around the active page.

Desktop sidebar

Control	Behavior
QuickMaths brand	Identifies the app; the subtitle shows the selected subject
Subject selector	Switches the visible subject and theme
Analog clock	Shows local time
Session timer	Counts the current open session
Profile total	Accumulates saved practice time for the learner
Dashboard	Progress overview, suggestion, recent attempts, and backup status
Mastery map	Prerequisite graph, lesson details, map scope, zoom, and personal Plan mode
Lessons	Theory, worked examples, applications, and recommended preparation
Mastery test	Complete authored assessment and saved response draft
Results	Grading, reflection, review, and mastery update
Lesson Depot	Public lesson-pack discovery, preview, installation, votes, and comments
Lesson Studio	Friendly authoring and reversible native-lesson improvements
Settings	Path mode, tutorial, curriculum, backups, storage, exports, and installed content
Profile badge	Opens Dashboard; the arrow returns to profile selection

Mobile bottom navigation

The bottom bar shows Home, Map, Learn, Test, Depot, and Settings. Lesson Studio is available from the Depot tab when it cannot fit as a separate item. The fixed navigation stays reachable while the page scrolls.

Subject theme

Each subject has a safe fixed color palette. The selected subject changes interface accents. In the combined map, each subject's node colors remain visible while mastery status uses text and status dots.

Agent Studio

The star-shaped nub opens Agent Studio. Its close button remains visible at different zoom levels and leaves the nub behind.

Agent Studio reports WebMCP availability, registered and failed tools, a suggested prompt, the available tool-name list, and agent-attributed activity. Your own button clicks do not appear as agent activity.

5. Learner Dashboard

Dashboard answers three questions: where am I, what should I do next, and is my work safely portable?

Page actions

- **Student guide** opens this PDF.
- **Save backup** downloads complete restorable state.
- **Open mastery map** opens the selected subject map.

Progress metrics

Cards summarize mastery states and lesson counts. Typical states are Ready, Learning, Proven, Mastered, Rusty, and Locked. Counts respect the current subject or attached curriculum.

Suggested next lesson

QuickMaths prioritizes due review, active learning, and ready work. The suggestion explains why the lesson matters and provides direct lesson or test actions. It is guidance, not a claim that every learner must follow the same order.

Recent attempts

Recent rows show lesson, date, score, review state, and mastery result. Open an attempt to inspect results, reflection, and any saved tutor feedback.

Backup recommendation

QuickMaths recommends a portable backup after meaningful new work, installed-content changes, review changes, curriculum changes, storage trouble, or an extended period without export. The reminder waits for a natural stopping point rather than interrupting every question.

Curriculum completion and contact

An attached curriculum can expose its educator contact and completion context. Email buttons open a draft in your mail app. QuickMaths does not send messages or attachments automatically.

6. Subjects and learning paths

Subject selector

The selector changes Dashboard, Map, Lessons, and Test context. A newly installed subject appears alongside Mathematics. Cross-subject prerequisites still refer to globally unique lesson IDs.

Hard path

Hard path enforces prerequisites. A connected mastery test remains locked until the required earlier lessons reach a proven state. The lesson page and map show what preparation is missing.

Open path

Open path keeps the same prerequisite connections as recommendations, but lessons and tests remain available. This is useful for review, placement, or learners entering with knowledge acquired elsewhere.

Curriculum-controlled mode

If an educator curriculum sets the path, the learner Settings controls are disabled. Loading a revised curriculum is the safe way to change educator-authored policy without silently editing it.

7. Mastery map

The map turns prerequisite structure and progress into one interactive canvas.

Current subject and all subjects

Current subject shows one focused graph. **All subjects** arranges installed subjects in labeled lanes, keeps each subject's colors, and draws cross-subject bridges.

Node status

Status	Meaning
Locked	Hard-path prerequisites are not yet proven
Ready	Available and not yet attempted
Learning	Attempted but not yet proven, or review/revision is pending
Proven	Strong recent evidence of understanding
Mastered	Sustained high mastery evidence
Rusty	Review is due after time has passed

Selecting a node updates the detail card and routed lesson without resetting the map's pan position.

Detail card

The card shows lesson name, subdomain, description, mastery score, latest score, confidence, prerequisites, unlocks, why the lesson matters, and available lesson/test actions. Hard path identifies unmet preparation; Open path labels it as guidance.

Zoom and movement on desktop

Use the plus and minus buttons or the mouse wheel while the pointer is over the map. Zoom ranges from a broad 10 percent overview to a readable close view. Drag empty space to pan horizontally and vertically. The page itself should not resize when the map zooms.

Zoom and movement on mobile

Pinch inside the map to zoom. Drag empty space to pan. The map viewport remains a stable window so pinch zoom changes the canvas rather than the entire page scale.

Jump to skill

The skill selector moves focus to a known lesson and updates the detail card. It is useful on large combined maps.

8. Personal Plan mode

Plan mode is a private working copy layered over the canonical mastery map. Turning it off restores the canonical view without deleting your saved plan.

Enter and leave Plan mode

Choose **Plan mode** in the map header. A toolbar and Plan details card appear while the map remains visible. Changes autosave with your learner profile and travel in full backups and Bridge checkpoints.

Desktop selection

- Click a node for a new selection.
- Ctrl-click adds or removes one node.
- Drag a rectangle across empty map space to select enclosed nodes.
- Hold Ctrl while box-selecting to add to the current selection.
- Drag a selected node to move the selected group.
- Shift-drag empty space to pan when selection gestures are active.

Mobile selection

- Long-press a node to select it.
- Long-press it again to deselect it.
- Long-press empty map space to clear the selection.
- Drag selected nodes to move them.
- Drag ordinary empty space to pan.

Custom paths

Select at least two lessons, choose **Custom path**, name the route, and choose an outline color. Selection order becomes path order. Bold connections and node outlines visualize the plan.

Custom paths do not change real prerequisites or mastery. They represent your intended study order, revision route, exam preparation, or project sequence.

Annotations

Choose **Annotation** to create a note connected to selected lessons, connected to a saved path, or free on the map when nothing is selected. Comment nodes can be dragged. Use annotations for goals, misconceptions, dates, resources, or questions. Do not store passwords or tokens in them.

Plan details and reset

The side card lists selection, saved paths, and annotations. Delete individual plan items or reset node positions for the visible scope. Subject layouts and the combined all-subject layout keep separate positions.

9. Lesson page

The lesson page is the study surface before and between assessments.

Lesson header

The header identifies subject, subdomain, lesson, description, preparation state, and available actions. A locked Hard-path test can still leave theory available for study.

Theory

Theory sections explain definitions, relationships, methods, notation, and edge cases authored for the lesson. Read them actively: pause to predict the next step, restate the idea, and identify assumptions.

Worked examples

Examples separate prompt, solution, and explanation. Try the prompt before reading the solution. Compare not only the final answer but the structure of the method.

Applications

Applications connect the skill to real decisions, modeling, geometry, another subject, or later mathematics. They explain why the lesson belongs in the map.

Recommended preparation and unlocks

Preparation links open prerequisite lessons. Unlocks show where the current skill leads. In Open path, these remain guidance; in Hard path, they also control assessment access.

Start mastery test

The button opens or resumes the lesson's complete assessment draft. QuickMaths uses authored scenarios rather than replacing substantial lessons with a generic five- or ten-question cap.

10. Mastery tests

Tests cover the authored lesson, including variations and edge cases supplied by native runtime templates or fixed validated lesson packs.

Saved drafts

Answers and shown work autosave while you type. Returning to the same test resumes the draft. Content updates can restart affected unfinished drafts so answers never cross between different question banks.

Question navigation

The page shows question number, total authored scenarios, prompt, response type, and any required work. Complete every visible authored scenario before submission.

Final-answer types

Questions may accept numeric answers, tolerance ranges, multiple-choice options, symbolic expressions, finite sets, rational expressions with domain exclusions, equation solutions, interval sets, exact text, or theorem conclusions. Follow the notation instructions in the prompt.

Finite sets are order-independent and ignore repeated entries. You can usually write $\{-2, 5\}$ or $x = -2$ or $x = 5$; use $\{\}$ or no solution for the empty set. Interval answers accept standard interval/union notation such as $(-\infty, -1] \cup (3, \infty)$, and many equivalent inequality forms. Infinity endpoints must always be open.

For a rational-expression question, the simplified formula and its excluded input values are graded together. A cancelled factor can leave a hole, so enter every original domain restriction in the separate **Excluded values** field even when that factor is no longer visible in the final formula.

Written explanations

A separate work box can capture your reasoning even when the final answer self-grades. Explanation text is saved for Results, reflection, and optional review.

Checked maths steps

Some advanced algebra questions require ordered steps. Put one equation, expression, or inequality per line. QuickMaths checks whether successive lines remain equivalent and whether the final line matches the answer. For inequalities, sign reversal and the full solution set matter.

This subsystem checks supported mathematical transformations. It is not a formal-proof judge.

Rational-equation ledger

Some rational equations open a guided work area for original restrictions, algebra steps, and candidate classifications. Record values excluded by the original denominators before clearing fractions. Then enter one algebraic statement per line and classify each candidate as valid, excluded, extraneous, repeated, or non-real. QuickMaths checks the restrictions, the algebraic progression, each candidate against the original equation, and whether the valid candidates match the submitted solution set.

Sign-chart workspace

Polynomial and rational inequalities can open a structured sign chart. Enter critical points, one test value and sign for each interval, selected intervals, endpoint decisions, and the final interval set. The page presents normal fields rather than JSON. QuickMaths reports errors by row, including a misplaced critical point, an incorrect test sign, a wrongly selected interval, or an endpoint that should be included or excluded.

Formal proof required

Proofs use ordinary text. No JSON, LaTeX, or magic keywords are required. One claim or reason per line is easiest to review.

The page shows a visible obligation checklist and accepted approaches. The short conclusion is graded separately; the proof is saved pending an obligation-by-obligation self, human, or agent review. Empty or extremely short work is blocked.

Required long response

Rubric questions show the criteria that will be reviewed. Address each criterion with evidence, calculations, sources, or reasoning appropriate to the prompt.

Submission boundary

Before submission, QuickMaths and its WebMCP tutor do not expose expected answers or hidden solution steps. Submit only when the final answers and required work are complete.

11. Results, reflection, and mastery

Grading results

Results compare your submitted final answers with allowed grading rules. After submission, authored solutions and explanations can be shown for study. Mistake tags help identify recurring patterns.

Work state

Each response reports whether required work was present, checked, pending review, passed, or needs revision. A correct final answer can coexist with a flawed proof or incomplete rubric response.

Structured reflection

Record confidence, felt difficulty, hint use, guessing, desire for more practice, confusing parts, and notes. Honest reflection helps distinguish understanding from luck or fragile recall.

Mastery update

The update combines assessment evidence, reflection, previous history, and required review state. It can move a lesson to Learning, Proven, Mastered, or later Rusty. QuickMaths does not claim mastery when a required proof or rubric review is still pending.

Self, human, and agent review

When allowed, the Results page can save a self review. Tutor-required work needs a human tutor or connected agent.

Proof review records a status and note for every obligation: satisfied, flawed, missing, or not applicable. Rubric review records points and evidence for every criterion. The reviewer also saves concise feedback, confidence, and one next step.

Review packet and email

Downloadable tutor summaries and review packets contain post-attempt answers, shown work, reflection, and review instructions. If the curriculum includes an educator contact, QuickMaths opens a prefilled email draft. You must attach and send the packet yourself.

Practice again

Retaking is always available as learning practice. Native lessons can generate fresh values while covering the same authored scenarios. QuickMaths curricula do not turn practice into one-shot unsupervised exams.

12. Lesson Depot

Lesson Depot discovers optional subjects and specialist tracks. Browsing, previewing, validating, and installing published packs do not require community authorization.

Search and filters

Search matches package name, subject, description, author, and tags. Filter published packages from roadmap concepts, choose subject, and sort by recommendation, recency, name, or community signal.

Package cards and themes

Cards use their designated subject colors. They identify version, author, lesson count, tags, compatibility, availability, and community signals.

Preview and installation safety

Published previews fetch a package, verify its catalog hash when browser crypto is available, and run local schema, size, graph, grader, theme, and content-shape validation. You see a preview and confirm installation.

Validation does not prove factual correctness. Review community content for quality, licensing, suitability, and accessibility.

Staged agent batches

An agent can stage one pack or an ordered batch for review. It cannot install. Settings presents a sequential queue: you separately install or skip every package. One approval never authorizes the next.

Upvotes and comments

Optional **Connect GitHub** authorizes the separate QuickMaths Community GitHub App for public Discussions on the QuickMaths repository. Connected humans can upvote and comment inside the app. These actions are public, use your GitHub identity, and are not WebMCP tools.

Community authorization is different from the fine-grained token used for private storage.

13. Lesson Studio for curious learners

Lesson Studio is available to everyone because explaining and authoring are powerful ways to learn. It can create new lessons or improve built-in QuickMaths lessons reversibly.

Friendly authoring flow

Choose a subject, write one or more lessons, add theory, worked examples, applications, prerequisites, and mastery questions, then validate and install. Question-mark controls provide hover, keyboard, and mobile-tap help.

Response and review design

Studio separates final-answer grading, shown work, and review. It explains ordinary answers, finite sets, rational formulas and exclusions, interval sets, checked maths steps, rational-equation ledgers, sign charts, formal proof obligations, and rubric responses with learner previews and editable examples.

Native improvements

Open a built-in lesson as an editable override while keeping its exact ID and subject. Completed progress and map identity stay attached. Affected unfinished tests restart. Settings can restore the original.

Human control

Validation reports issues; it does not silently install or publish. Installing requires confirmation. Lesson Depot publication opens a separate human contribution workflow.

14. Learning with an agent

WebMCP lets a compatible browser agent inspect the same visible QuickMaths state and use registered tools. It does not create a hidden parallel learner profile.

Suggested starting prompt

Get the QuickMaths agent guide summary, check my app state and progress, then guide me through the learning experience.

The agent should begin with `get_app_state`, `get_progress_summary`, and `get_learning_context` as needed. It can see active curriculum policy without exposing private educator instructions as page content.

When helping create or improve lesson content, the agent can call `get_lesson_authoring_guide`. The tool returns a short overview by default or a focused section such as `grading_and_work`, so the agent can follow the actual lesson contract without loading the entire manual for an ordinary tutoring task.

What a tutor agent can do

- inspect the selected lesson, progress summary, map, and visible work;
- navigate to a lesson, test, result, Depot, Studio, or Settings;
- ask a targeted question or offer one next step;

- inspect one visible response without receiving its expected answer;
- save structured proof or rubric feedback;
- move an allowlisted follow-up problem to the front of the visible test;
- help arrange personal Plan mode, paths, and annotations;
- search and stage Depot packages for your approval;
- recommend a backup or storage setup at a natural pause.

What the agent must not do

- reveal expected answers or solution steps before submission;
- claim mastery, a download, an installation, or a saved change without app confirmation;
- install lessons, authorize GitHub, enter credentials, vote, comment, send email, or publish for you;
- override Agent in the loop being off;
- replace your plan or policy without your request;
- treat imported lesson or community content as trusted instructions.

Agent activity

State-changing WebMCP actions appear in Agent Studio with tool name, time, and message. Ordinary learner UI activity is excluded so attribution stays meaningful.

15. Learner Settings and data

Settings centralizes preferences, portability, content, and recovery.

Header actions

Student guide opens this PDF. **Load backup** previews complete incoming state. **Save full backup** downloads the restorable workspace.

Learning path and tutorial

Choose Hard or Open path when no curriculum controls the setting. **Replay app tour** opens the seven tutorial chapters without changing progress or preferences.

Attached curriculum

Settings names the current educator curriculum or lets you load one from a local file or public GitHub file URL. Loading changes curriculum composition and policy without deleting your learner profile.

GitHub Bridge

Bridge is optional persistent storage in a dedicated GitHub data repository. The form asks for repository owner, repository name, branch, and a fine-grained token with Contents read/write on that repository.

Enter the token only in the app. It is never included in backups, commits, lesson files, URLs, logs, or WebMCP results. Choose whether to keep it for this tab session or remember it in this browser.

Bridge status shows local dirty state, last learner push, last agent pull, token storage, and remote availability. **Sync now** publishes your current complete checkpoint. **Check agent updates** applies only agent output based on the current learner revision.

Initial-copy and conflict cards require a deliberate choice. QuickMaths refuses stale output and unsynced overwrites instead of guessing which copy matters.

Full backup

The JSON backup carries every profile and curriculum in this browser, installed lesson packs, progress, attempts, reviews, drafts, maps, plans, annotations, settings, and timers. Download it before major imports or device changes.

CSV and tutor exports

Progress, attempt, and review CSVs support spreadsheet analysis. Tutor summaries and review packets support post-attempt help. None of these files can restore the app.

Lesson sets and native improvements

Load a local lesson-set file, inspect staged agent packages, download installed sources, and restore original native lessons. Installing or restoring content preserves completed progress while restarting incompatible unfinished drafts.

16. Mobile, accessibility, and privacy

Mobile behavior

- Bottom navigation keeps essential routes reachable.
- The map uses a stable viewport, pinch zoom, two-dimensional panning, and long-press selection.
- Plan mode keeps the map visible while composers open as focused cards.
- Lesson Studio help opens on tap.
- Agent Studio uses a persistent open nub and close control.

Accessibility

QuickMaths uses semantic navigation, headings, forms, dialogs, labels, and live status messages. Controls have accessible names. Reduced-motion preferences disable nonessential movement. Subject colors are paired with text, structure, and status indicators.

Privacy boundaries

Treat answers, shown work, reflection, notes, repository details, and imported content as private. Do not paste storage tokens into chat. Community votes and comments are public. Curriculum agent instructions are visible to a compatible agent but not rendered as learner page content.

17. A strong learning routine

1. Check Dashboard for due review and suggested work.
2. Open the map and inspect prerequisites, unlocks, and educator paths.
3. Read the lesson actively and attempt examples before reading solutions.
4. Complete the full mastery test, including required work.
5. Reflect honestly before seeing mastery update.
6. Review mistakes, proof obligations, rubric feedback, and solution explanations.
7. Add a Plan mode annotation or path when a concrete future action will help.
8. Retake with fresh scenarios after revision rather than memorizing an answer.
9. Ask an agent or human tutor for one targeted next step when stuck.
10. Save a full backup or sync Bridge at a natural stopping point.

18. Troubleshooting and quick reference

A test is locked

Check Hard path prerequisites on the map and lesson page. If you already know the material from elsewhere and no curriculum controls the mode, choose Open path in Settings.

The selected map card is wrong

Select the node once and confirm the highlighted node and detail title match. The app preserves pan position after selection. Reload the latest deployed version if an old cache is still open.

Map zoom changes the whole page

Use pinch or wheel inside the map viewport. The current app isolates canvas zoom from page scale. Refresh if a stale release is cached.

A proof has the right conclusion but remains Learning

The conclusion and proof are separate judgments. Open Results and complete the required obligation review with an allowed reviewer.

A staged batch did not install everything

That is intentional. Review, install, or skip each package separately in Settings.

A package is installed but missing from an educator curriculum

Installation adds it to the browser library. The educator must enable the additive pack for that curriculum, then export a revised curriculum.

The agent cannot tutor

The attached curriculum may have Agent in the loop turned off. That policy deliberately blocks tutoring and learner-work tools.

GitHub sync reports a conflict

Download a backup if needed, read which complete copy is newer, and choose deliberately. Do not retry blindly. GitHub repository history remains a recovery aid.

WebMCP tools are unavailable

Open QuickMaths in a compatible ChatGPT or Codex browser. Agent Studio names registration status and any individual failures.

Essential links

App and source: <https://quickmathematics.github.io/QuickMaths/> and <https://github.com/QuickMathematics/QuickMaths>

Technical guides: https://quickmathematics.github.io/QuickMaths/CUSTOM_LESSON_SETS.md and <https://quickmathematics.github.io/QuickMaths/bridge-guide.html>